Supplement: Verifier-Guided Constraint Projection

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A Implementation details of the verifier

The pairwise overlap matrix $A \in \mathbb{R}^{N \times N}$ is computed as follows (PyTorch pseudocode):

```
# centers: (N, 2), sizes: (N, 2)
cx, cy = centers[:, 0], centers[:, 1]
sw, sh = sizes[:, 0], sizes[:, 1]

# pairwise AABB
ovx = torch.clamp_min(
    torch.minimum(cx[:, None] + sw[:, None]/2, cx[None, :] + sw[None, :]/2) -
    torch.maximum(cx[:, None] - sw[:, None]/2, cx[None, :] - sw[None, :]/2),
    0,
)
ovy = torch.clamp_min(
    torch.minimum(cy[:, None] + sh[:, None]/2, cy[None, :] + sh[None, :]/2) -
    torch.maximum(cy[:, None] - sh[:, None]/2, cy[None, :] - sh[None, :]/2),
    0,
)
A = ovx * ovy
A.fill_diagonal_(0)  # exclude self-overlap
```

Boundary violation for each macro along axis $a \in \{x, y\}$ and edge $e \in \{-, +\}$:

$$b_i^{a,-} = \max(0, s_i^a/2 - c_i^a), \qquad b_i^{a,+} = \max(0, c_i^a + s_i^a/2 - W_a), \tag{1}$$

with $b_i = \sum_a (b_i^{a,-} + b_i^{a,+})$.

B Neural repair operator: full training details

Architecture. 3-layer GATv2 (heads=4, hidden=128), scale-invariant input normalization (centers and severity are divided by $\max(W, H)$). Node features: $[\text{severity}, \text{boundary}, \text{n_overlappers}, s^w, s^h]$. Edge features: 4-dim pin offsets (from ChipDiffusion's edge_attr).

Loss. L2 on the predicted displacement against a target defined by the variant (see below).

Variants.

- **Synthetic teacher**: hand-crafted operator applied to synthetic placements from ChipDiffusion's data generator. 2500 training graphs, 200 val.
- **Real teacher**: hand-crafted operator on diffusion-sampled ISPD2005 placements (3 samples per circuit $\times$ 1–3 circuits = 3–9 samples depending on seed).

- **Data augmentation**: per-sample $\times 50$ random Gaussian perturbations of the baseline.
- **Trajectory distillation**: (x_k, x_{k+1}) pairs from every step of the hand-crafted operator; 30 steps $\times$ 3 samples $= 90$ pairs.
- **Ground-truth target**: ISPD2005 industrial `.pl` placements as target.
- **Residual learning**: train to predict $x_{GT} - x_{\text{hand}}$ after hand-crafted convergence; apply as residual scale $s \in \{0.01, 0.05, 0.1, 0.2\}$.

Training. Adam, lr 10^{-3}, 100 epochs, batch 1 (per-graph), early stop on val loss.

Findings. Best val loss was 0.0037 (trajectory distillation), but at inference on ISPD2005 the best variant (residual $s = 0.1$) improved violations on 1 of 6 circuits while degrading HPWL. Detailed per-circuit results are available in `results/neural_vsr/eval_*.json`.

C Extended Pareto results

Full per-seed per-λ results are in `results/ispd2005/pareto_3seed_6w.json`. The 3-seed standard deviation for Δviol is $< 4\%$ on 5 of 6 circuits and $< 8\%$ on adaptec2. The standard deviation for ΔHPWL is circuit-dependent: under 15% for adaptec1–4 and bigblue3, and up to 20% on bigblue1 where baseline HPWL is very small and therefore percent-changes are noisier.

D Convergence curves

Figure 1 (attached) shows per-iteration (v_t, h_t) trajectories for $T = 300$ iterations on adaptec1 and bigblue1 at $\lambda = 2$. Violations decrease near-monotonically; HPWL changes non-monotonically in the first ~ 30 iterations (as the repulsive force initially pushes connected macros apart) before the attractive term pulls them back. The plateau is reached by $T \approx 100$, consistent with the num_steps ablation.

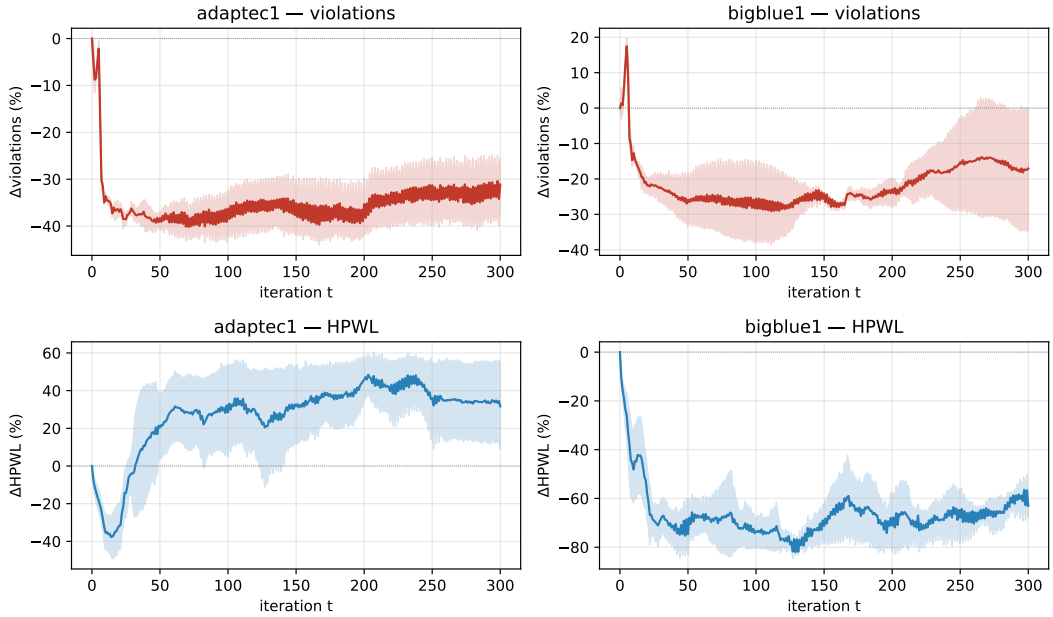

Figure 1: Per-iteration convergence for VSR on adaptec1 and bigblue1, 3 seeds each. Top row: violations vs. iteration. Bottom row: HPWL vs. iteration. Shaded band = seed std.

E Full ablations and extra figures

Number of steps T (Fig. 2). Violations plateau near $T = 100$; going to $T = 500$ gains $< 2\%$ on all circuits and HPWL is stable.

Step size η (Fig. 3). $\eta = 0.3$ (default) is the robust middle ground. $\eta = 1.0$ gains marginal violation reduction at higher HPWL variance.

Selective vs. global (Tab. 1). Identical on ISPD2005 raw drafts because every macro violates.

Intra-sampling start timestep (Fig. 4). For bigblue3, $t = 0.3$ achieves the best HPWL (-16%); $t = 0.5$ gives $+9\%$. For adaptec3 the curve is flatter, confirming intra's circuit-dependence.

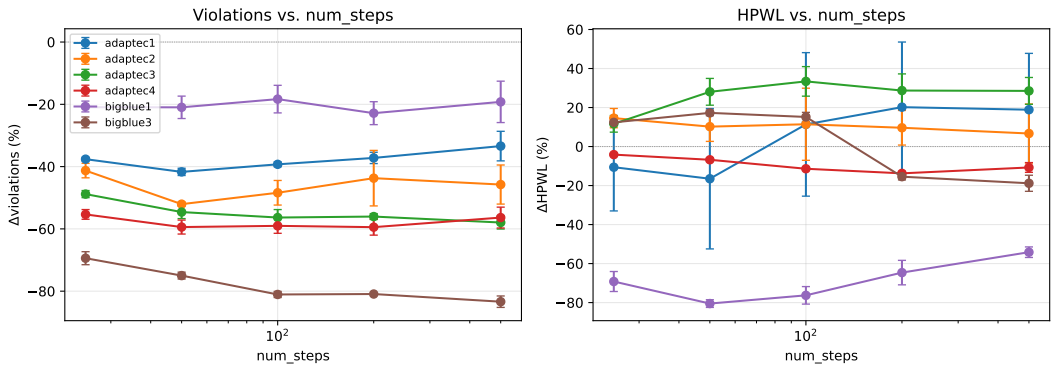

Figure 2: Number of repair steps ablation.

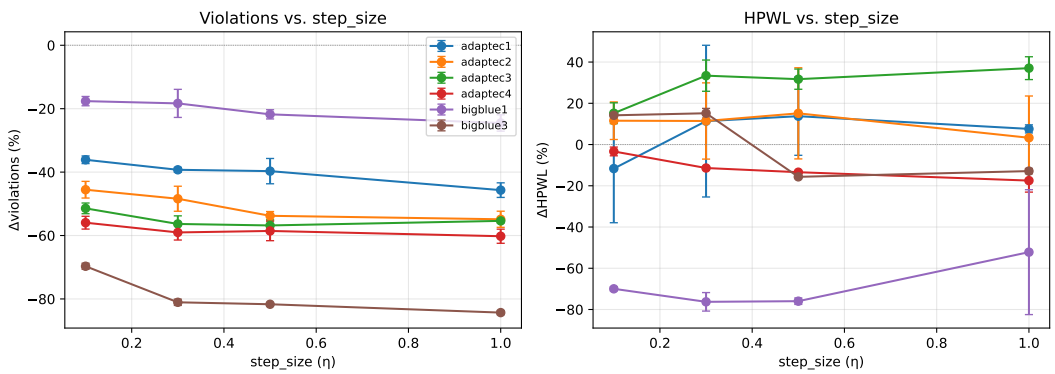

Figure 3: Step size ablation.

F Reproducibility

All code, configs, checkpoints, and result JSONs are released. A single command reproduces all main-paper numbers:

```
python scripts/run_all_experiments.py --config configs/paper_main.yaml
```

on an A800 80GB GPU (or equivalent). The full sweep takes ~ 20 minutes.

Circuit	Δviol on	Δviol off	ΔHPWL on	ΔHPWL off
adaptec1	−39.3	−39.3	+11.4	+11.4
adaptec2	−48.4	−48.4	+11.4	+11.4
adaptec3	−56.4	−56.4	+33.4	+33.4
adaptec4	−59.0	−59.0	−11.4	−11.4
bigblue1	−18.3	−18.3	−76.3	−76.3
bigblue3	−81.1	−81.1	+15.2	+15.2

Table 1: Selective vs. global repair.

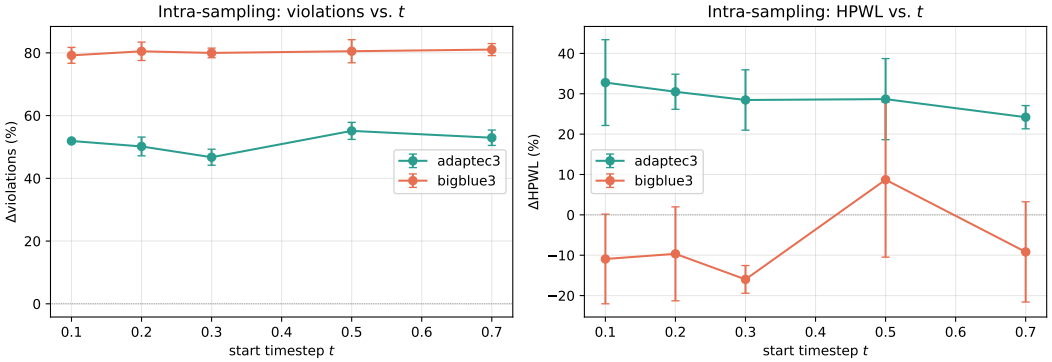

Figure 4: Intra-sampling start timestep sweep.